COL750 Assignment-2

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Part-1

Problem 1.1

(a)

The color of Sabhya's hat is red.

Justification: Considering Arunabh does not know the color of his hat, the possible colors of the hats of Dinu and Sabhya can be represented by the set $\{rr, wr, rw\}$ (first for Dinu, second for Sabhya). This becomes *common knowledge* after Arunabh's announcement. Since Dinu can see Sabhya's hat, if Sabhya's hat were white then Dinu would know his own hat is red. Dinu not knowing his hat color forces the condition that Sabhya's hat must be red.

(b)

Initial Kripke Structure: Define $M = \langle S, \sim, V \rangle$ where

$$\begin{gathered}
S = \{rrr, rrw, rwr, rww, wrr, wrw, wwr\} \\
\text{Agents} = \{a, d, s\} \\
\sim_a (rrr) = \{rrr, wrr\} = \sim_a (wrr) \\
\sim_a (rrw) = \{rrw, wrw\} = \sim_a (wrw) \\
\sim_a (rwr) = \{rwr, wwr\} = \sim_a (wwr) \\
\sim_a (rww) = \{rww\} \qquad (1.1(b1)) \\
\text{(Similarly } \sim_d, \sim_s \text{ are defined for the other agents.)} \\
V(r_a) = \{rrr, rwr, rrw, rww\}, \quad V(w_a) = S \setminus V(r_a) \\
V(r_d) = \{rrr, wrr, rrw, wrw\}, \quad V(w_d) = S \setminus V(r_d) \\
V(r_s) = \{rrr, wrr, rwr, wwr\}, \quad V(w_s) = S \setminus V(r_s)
\end{gathered}$$

We define the announcements as follows: ϕ_a means "Arunabh cannot determine his hat color", and ϕ_d means "Dinu cannot determine his hat color". We encode them as

$$\phi_i = \neg K_i r_i \wedge \neg K_i w_i, \qquad \forall i \in \{a, d\}.$$

Now we see how the announcements restrict the Kripke structure.

After announcement ϕ_a: The restricted Kripke structure of M is $M|_{\phi_a}$ where

$$S^{(a)} \coloneqq [\![\phi_a]\!] \coloneqq \{s \in S : M, s \models \phi_a\}$$
$$\forall i \in \{a, d, s\} : \sim_i^{(a)} \coloneqq \sim_i \cap ([\![\phi_a]\!] \times [\![\phi_a]\!])$$
$$V^{(a)}(p) \coloneqq V(p) \cap [\![\phi_a]\!]$$

For any world $s \in S$,

$$\begin{aligned} M, s \models \phi_a &\iff M, s \models \neg K_a r_a \wedge \neg K_a w_a \\ &\iff \neg(M, s \models K_a r_a) \wedge \neg(M, s \models K_a w_a) \\ &\iff \neg(\forall t \in \sim_a (s) : t \in V(r_a)) \wedge \neg(\forall t \in \sim_a (s) : t \in V(w_a)) \end{aligned} \tag{1.1(b2)}$$

From the definition of $\sim_a$ in (1.1(b1)) , we observe:

$$\begin{aligned} &\forall s \in S \setminus \{rww\} : \exists t_1, t_2 \in \sim_a (s) : t_1 \notin V(r_a) \wedge t_2 \notin V(w_a) \\ \iff &\forall s \in S \setminus \{rww\} : \neg(\forall t \in \sim_a (s) : t \in V(r_a)) \wedge \neg(\forall t \in \sim_a (s) : t \in V(w_a)) \\ \iff &\forall s \in S \setminus \{rww\} : M, s \models \neg K_a r_a \wedge \neg K_a w_a \end{aligned} \tag{1.1(b3)}$$

Thus $S^{(a)} = [\![\phi_a]\!] = S \setminus \{rww\}$. The equivalence relations and valuation are restricted accordingly.

Intuitive verification: Since there are only two white hats, if Dinu and Sabhya both had white hats, Arunabh would definitely know his hat is red. Hence the world rww is eliminated after ϕ_a. In all other worlds, Arunabh lacks sufficient information to know his hat color, so they survive.

After announcement ϕ_d: Now restrict $M|_{\phi_a}$ further:

$$S^{(d)} \coloneqq [\![\phi_d]\!] \coloneqq \{s \in S^{(a)} : M|_{\phi_a}, s \models \phi_d\}$$
$$\forall i : \sim_i^{(d)} \coloneqq \sim_i^{(a)} \cap ([\![\phi_d]\!] \times [\![\phi_d]\!])$$
$$V^{(d)}(p) \coloneqq V^{(a)}(p) \cap [\![\phi_d]\!]$$

For $s \in S^{(a)}$,

$$M|_{\phi_a}, s \models \phi_d \iff \neg(\forall t \in \sim_d^{(a)} (s) : t \in V^{(a)}(r_d)) \wedge \neg(\forall t \in \sim_d^{(a)} (s) : t \in V^{(a)}(w_d)) \tag{1.1(b4)}$$

The restricted equivalence relation $\sim_d^{(a)}$ is:

$$\begin{aligned} \sim_d^{(a)} (rrr) &= \{rrr, rwr\} = \sim_d^{(a)} (rwr) \\ \sim_d^{(a)} (wrr) &= \{wrr, wwr\} = \sim_d^{(a)} (wwr) \\ \sim_d^{(a)} (wrw) &= \{wrw\} \\ \sim_d^{(a)} (rrw) &= \{rrw\} \end{aligned} \tag{1.1(b5)}$$

We observe:

$$\forall s \in S^{(a)} \setminus \{wrw, rrw\} : \exists t_1, t_2 \in \sim_d^{(a)} (s) : t_1 \notin V^{(a)}(r_d) \wedge t_2 \notin V^{(a)}(w_d)$$
$$\iff \forall s \in S^{(a)} \setminus \{wrw, rrw\} : M, s \models \neg K_d r_d \wedge \neg K_d w_d \qquad (1.1(b6))$$

Hence $S^{(d)} = S^{(a)} \setminus \{wrw, rrw\}$.

Intuitive verification: Because there are only two white hats, if Arunabh and Sabhya both had white hats (world wrw), Dinu would know his hat is red. Similarly, if Arunabh has red and Sabhya has white, the two possible worlds are $\{rrw, rww\}$; but rww was already eliminated, so Dinu would know his hat is red. Therefore both wrw and rrw are eliminated after ϕ_d. The remaining worlds survive because Dinu cannot deduce his hat color.

(c)

Initial Kripke Structure (Sabhya blindfolded): $M = \langle S, \sim, V \rangle$ with the same set S and valuations as in (b), but now Sabhya is blindfolded, so he cannot distinguish any worlds:

$$S = \{rrr, rrw, rwr, rww, wrr, wrw, wwr\}$$
$$\text{Agents} = \{a, d, s\}$$
$$\sim_s = S \times S$$
$$(\sim_a, \sim_d \text{ and V are as in (b).})$$

Problem 1.2

(a)

Each announcement by Tandon becomes common knowledge. Consider the repeated announcement: *"Does any of you know whether you have mud on your own forehead?"*. If no child says "yes" after the first repetition, then all children know that there are **more than one** muddy child. The justification is simple: each child sees the others' foreheads. If there were exactly one muddy child, that child would see all others clean and would therefore know his own muddiness immediately after the first announcement *"At least one of you has mud on your forehead"*. Since he does not say "yes", the situation with a single muddy child is impossible.

Similarly, if no child says "yes" after the second repetition, all children deduce there are **more than two** muddy children. By induction, after the i-th repetition with no "yes", the children know there must be **more than** i muddy children. Conversely, if a child knows there are more than $i-1$ muddy children (from the previous round's silence) and sees only $i-2$ muddy children among the others, he deduces he himself must be muddy. Hence, after k repetitions, every child knows whether he has mud on his forehead, because the set of possible worlds will become empty.

Initial Kripke structure: Define $M = \langle S, \sim, V \rangle$ where

$$S = \{0,1\}^n,$$
$$\text{Agents} = \{c_i \mid 1 \le i \le n\},$$
$$\forall i \in \{1,\ldots,n\} : \sim_{c_i} (l1r) = \{l1r, l0r\} = \sim_{c_i} (l0r) \quad \text{and } \forall l \in \{0,1\}^{i-1}, \forall r \in \{0,1\}^{n-i}$$
$$\forall i \in \{1,\ldots,n\} : V(0_{c_i}) = \{l0r : \forall l \in \{0,1\}^{i-1}, \forall r \in \{0,1\}^{n-i}\}, \quad V(1_{c_i}) = S \setminus V(0_{c_i}). \tag{1.2(a1)}$$

The Kripke structure shows all possible worlds as n-length bit strings, where the i-th bit 0 means child c_i is clean, otherwise muddy. The equivalence relation $\sim$ reflects the fact that each child can see the others' foreheads but not their own. The proposition 0_{c_i} represents that child c_i is clean, and $V(0_{c_i})$ is the set of worlds where that holds.

Now we examine how each announcement by Tandon restricts the Kripke structure M. Let ϕ_0 denote the announcement *"At least one of you has mud on your forehead"*, and let ϕ denote the announcement *"Does any of you know whether you have mud on your own forehead?"*. We write ϕ_i (with $i \ge 1$) for the i-th repetition of ϕ.

After the announcement ϕ_0: We encode ϕ_0 as $\phi_0 = \bigvee_{i=1}^{n} 1_{ci}$. The restricted Kripke structure is $M^{(0)} = M|_{\phi_0}$ where

$$S^{(0)} \coloneqq [\![\phi_0]\!] \coloneqq \{s \in S : M, s \models \phi_0\},$$
$$\forall i \in \{1,\ldots,n\} : \sim_{c_i}^{(0)} \coloneqq \sim_{c_i} \cap ([\![\phi_0]\!] \times [\![\phi_0]\!]),$$
$$V^{(0)}(p) \coloneqq V(p) \cap [\![\phi_0]\!]. \tag{1.2(a2)}$$

For any world $s \in S$,

$$\begin{aligned} M, s \models \phi_0 &\iff M, s \models \bigvee_{i=1}^{n} 1_{ci} \\ &\iff \bigvee_{i=1}^{n} M, s \models 1_{c_i} \\ &\iff \bigvee_{i=1}^{n} s \in V(1_{c_i}). \end{aligned} \tag{1.2(a3)}$$

Except the world $\{0\}^n$, all worlds satisfy (1.2(a3)) ; hence only $\{0\}^n$ is eliminated from $M^{(0)}$. Thus $S^{(0)} = S \setminus \{0\}^n$, and the relations and valuation are restricted accordingly.

Intuitive verification: The announcement itself forces the condition that there is at least one muddy child, so the world with no muddy child becomes impossible. All other worlds have at least one muddy child and therefore survive.

After the announcement ϕ_i for $i \geq 1$: By construction $\phi_i = \phi$ for all $i \geq 1$; the index i denotes the repetition round. We encode ϕ as $\phi = \bigwedge_{j=1}^{n} \neg K_{cj} 1_{cj} \wedge \neg K_{cj} 0_{cj}$. This formula only allows worlds where no child knows whether he is muddy.

After the i-th repetition, the restricted Kripke structure is $M^{(i)} = M^{(i-1)}|_{\phi_i} = \langle S^{(i)}, \sim^{(i)}, V^{(i)} \rangle$ with

$$
\begin{aligned}
S^{(i)} &\coloneqq [\![\phi_i]\!] \coloneqq \{s \in S^{(i-1)} : M^{(i-1)}, s \models \phi_i\}, \\
\forall i \in \{1, \ldots, n\} &: \sim_{c_i}^{(i)} \coloneqq \sim_{c_i}^{(i-1)} \cap ([\![\phi_i]\!] \times [\![\phi_i]\!]), \\
V^{(i)}(p) &\coloneqq V^{(i-1)}(p) \cap [\![\phi_i]\!].
\end{aligned} \tag{1.2(a4)}
$$

For any world $s \in S^{(i-1)}$ and $i = 1, 2, \ldots,$

$$
\begin{aligned}
M^{(i-1)}, s \models \phi_i &\iff M^{(i-1)}, s \models \bigwedge_{j=1}^{n} \neg K_{cj} 1_{cj} \wedge \neg K_{cj} 0_{cj} \\
&\iff \bigwedge_{j=1}^{n} \big(M^{(i-1)}, s \models \neg K_{c_j} 1_{c_j} \ \wedge \ M^{(i-1)}, s \models \neg K_{c_j} 0_{c_j} \big) \\
&\iff \bigwedge_{j=1}^{n} \big(\neg (M^{(i-1)}, s \models K_{c_j} 1_{c_j}) \ \wedge \ \neg (M^{(i-1)}, s \models K_{c_j} 0_{c_j}) \big) \\
\iff \bigwedge_{j=1}^{n} \big(&\neg (\forall t \in \sim_{c_j}^{(i-1)} (s) : t \in V^{(i-1)}(1_{c_j})) \wedge \\
&\neg (\forall t \in \sim_{c_j}^{(i-1)} (s) : t \in V^{(i-1)}(0_{c_j})) \big).
\end{aligned} \tag{1.2(a5)}
$$

Case $i = 1$ (first repetition): Let $m_{j1} = \{l1r : \forall l \in \{0\}^{j-1}, \forall r \in \{0\}^{n-j}\}$ be the set of worlds where only child c_j is muddy, and $m_1 = \bigcup_{j=1}^{n} m_{j1}$ the set of worlds with exactly one muddy child. Observe that **for all** j,

$$
\forall l1r \in m_{j1} : \quad \sim_{c_j}^{(0)} (l1r) = \{l1r\} \quad (\text{since } l0r \notin S^{(0)}), \tag{1.2(a6)}
$$

$$
\begin{aligned}
&\forall l1r, l0r \in S^{(0)} \setminus m_1 \text{ and, } \forall l \in \{0,1\}^{j-1}, \forall r \in \{0,1\}^{n-j} : \\
&\sim_{c_j}^{(0)} (l1r) = \{l1r, l0r\} = \sim_{c_j}^{(0)} (l0r).
\end{aligned} \tag{1.2(a7)}
$$

From (1.2(a5)) and (1.2(a6)) , for any child c_j and any $s \in m_{j1}$, we have $M^{(0)}, s \models K_{c_j} 1_{c_j} (\because l1r \in 1_{c_j})$ and $M^{(0)}, s \not\models K_{c_j} 0_{c_j} (\because l1r \notin V^{(0)}(0_{c_j}))$. Hence all worlds in m_{j1} fail to satisfy ϕ_1 and are eliminated from $M^{(1)}$. This elimination is possible only because the world $\{0\}^n$ was already removed from $S^{(0)}$, which makes the equivalence classes of worlds in m_{j1} singletons. For any world $s \in S^{(0)} \setminus m_1$, from (1.2(a7)) ($l1r \notin V^{(0)}(0_{c_j})$ and $l0r \notin V^{(0)}(1_{c_j})$), where "$l1r$" and "$l0r$" belongs to $\sim_{c_j}^{(0)} (s)$. Hence, we obtain

$$
\bigwedge_{j=1}^{n} \big(M^{(0)}, s \not\models K_{c_j} 1_{c_j} \ \wedge \ M^{(0)}, s \not\models K_{c_j} 0_{c_j} \big),
$$

so they satisfy ϕ_1 and survive. Consequently, $S^{(1)} = S^{(0)} \setminus m_1$, i.e., all worlds with exactly one muddy child are removed after the first repetition.

Case $i = 2$ (second repetition): By the same reasoning as in case $i = 1$, because worlds with exactly one muddy child were eliminated after ϕ_1, the equivalence classes for worlds with exactly two muddy children become singletons. Therefore, after the second repetition ϕ_2, all worlds with exactly two muddy children are eliminated.

We can frame this as a proof by induction: after the i-th repetition of ϕ, only worlds with **more than i muddy children** remain in $M^{(i)}$. Hence, after k repetitions, all children know their own muddy status, because the set of possible worlds will become empty.

Intuitive verification: If there were exactly one muddy child, then after the announcement ϕ_0, that child would know he is muddy because he sees all others clean. Similarly, if no child says "yes" after the first repetition, then any muddy child must see at least one other muddy child; otherwise he would have deduced his own muddiness. This reasoning repeats: after the i-th repetition with no "yes", each muddy child sees at least i other muddy children, so he still cannot be certain. But once the number of repetitions reaches the actual number of muddy children, every muddy child sees only $k - 1$ muddy others and deduces he must be muddy. Hence, after k repetitions, all k muddy children know they are muddy, because the set of possible worlds will become empty.

(b)

Initial Kripke structure: Define $M = \langle S, \sim, V \rangle$ with the same set S and valuations as in (a), but now all children have bandages over their eyes, so they cannot distinguish any worlds:

$$
\begin{gathered}
S = \{0,1\}^n, \\
\text{Agents} = \{c_i \mid 1 \leq i \leq n\}, \\
\forall i \in \{1, \dots, n\} : \sim_{c_i} = S \times S, \\
\forall i \in \{1, \dots, n\} : V(0_{c_i}) = \{l0r : \forall l \in \{0,1\}^{i-1}, \forall r \in \{0,1\}^{n-i}\}, \quad V(1_{c_i}) = S \setminus V(0_{c_i}).
\end{gathered}
$$

After announcement ϕ_0: As shown in part (a) (1.2(a3)) , after the announcement ϕ_0, the restricted Kripke structure $M^{(0)} = M|_{\phi_0}$ satisfies

$$
\begin{gathered}
S^{(0)} = S \setminus \{0\}^n, \\
\forall i \in \{1, \dots, n\} : \sim_{c_i}^{(0)} = S^{(0)} \times S^{(0)}.
\end{gathered}
\tag{1.2(b1)}
$$

The valuation function is restricted accordingly, following the definition from part (a) (1.2(a2)) .

After announcement ϕ_i for $i \geq 1$: After the i-th repetition, the restricted Kripke structure is $M^{(i)} = M^{(i-1)}|_{\phi_i} = \langle S^{(i)}, \sim^{(i)}, V^{(i)} \rangle$, defined similarly to part (a) (1.2(a4)) .

Case $i = 1$**:** Since the children cannot see, using formula (1.2(a4)) and the equivalence relation (1.2(b1)) , for any world $s \in S^{(0)}$:

$$
\begin{aligned}
&\forall j \in \{1, \dots, n\} : \exists t_1, t_2 \in \sim_{c_j}^{(0)} (s) : t_1 \notin V^{(0)}(1_{c_j}) \ \wedge \ t_2 \notin V^{(0)}(0_{c_j}) \\
&\because (t_1 \in S^{(0)} \setminus V^{(0)}(1_{c_j}) \text{ and } t_2 \in S^{(0)} \setminus V^{(0)}(0_{c_j})) \\
\iff &\forall j \in \{1, \dots, n\} : \neg(M^{(0)}, s \models K_{c_j} 1_{c_j}) \ \wedge \ \neg(M^{(0)}, s \models K_{c_j} 0_{c_j}) \\
\iff &\forall j \in \{1, \dots, n\} : M^{(0)}, s \models \neg K_{c_j} 1_{c_j} \wedge \neg K_{c_j} 0_{c_j} \\
\iff &\bigwedge_{j=1}^{n} M^{(0)}, s \models \neg K_{c_j} 1_{c_j} \wedge \neg K_{c_j} 0_{c_j}.
\end{aligned}
$$

Thus every world in $S^{(0)}$ satisfies the announcement ϕ_1, so $S^{(1)} := [\![\phi_1]\!] = S^{(0)}$. This reflects the fact that in every world of $S^{(0)}$, no child knows his muddy status. In fact, $M^{(1)} = M^{(0)}$.

Case $i = 2, \dots$**:** The same pattern holds for all further repetitions of ϕ; (because the structure $M^{(i-1)}$ and announcement ϕ is same for all future repetition, so we'll get same result) the Kripke structure remains unchanged after each round. Hence,

$$\forall i \in \{2, \dots\} : M^{(i)} = M^{(i-1)} = M^{(0)}.$$

Intuitive verification: Since no child can see the status of others, they lack the necessary information to determine their own muddiness. Their only knowledge about the actual world is that there is at least one muddy child (from ϕ_0). Consequently, they will never learn their own muddy status, even after infinitely many repetitions of ϕ.

(c)

Initial Kripke structure: We use the same initial Kripke structure as in M defined in (1.2(a1)) . To encode Tandon's statement, we define an action model which captures the inattentiveness of *child 1*. Define $M^a = \langle S^a, \sim^a, prec \rangle$ such that

$$
\begin{aligned}
S^a &= \{A, B\}, \\
\sim^a &= \{(s, s) : s \in S^a\}, \\
prec(A) &= \bigvee_{j=1}^{n} 1_{c_j}, \\
prec(B) &= \text{True}.
\end{aligned}
$$

In the action model M^a, A represents the announcement *"At least one of you has mud on your forehead"* and B represents a vacuous statement (always true).

After Tandon's statement: The modified Kripke structure after Tandon's statement is the product update of M and M^a: $M'^{(0)} = M \otimes M^a = \langle S'^{(0)}, \sim'^{(0)}, V'^{(0)} \rangle$, where

$$S'^{(0)} = \{(s,a) : s \in S,\ a \in S^a,\ M, s \models prec(a)\},$$
$$\text{Agents} = \{c_j : j \in \{1, 2, \ldots, n\}\},$$
$$\sim'^{(0)}_{c_1} = \{((s_1, a_1), (s_2, a_2)) : \forall s_1 \in S,\ \forall s_2 \in \sim_{c_1} (s_1),\ \forall a_1, a_2 \in S^a\}, \tag{1.2(c1)}$$
$$\forall j \in \{2, 3, \ldots, n\} : \sim'^{(0)}_{c_j} = \{((s_1, a), (s_2, a)) : \forall s_1 \in S,\ \forall s_2 \in \sim_{c_1} (s_1),\ \forall a \in S^a\}, \tag{1.2(c2)}$$
$$V'^{(0)}(p) = \{(s, a) : \forall s \in V(p),\ \forall a \in S^a\}.$$

The equivalence relation for child c_1 differs from that of other children because c_1 cannot distinguish between A and B, whereas the others can. The restricted structure does not contain the world $(\{0\}^n, A)$ because, by construction of the product update, $M, \{0\}^n \not\models prec(A)$, while $prec(B) = \text{True}$; thus $(s, B) \in S'$ for all $s \in S$.

(i) For $n = 2$, the Kripke structure is as follows.

Before Tandon's statement M:

$$\begin{aligned}
&S = \{00, 01, 10, 11\},\\
&\sim_{c_1} (00) = \{00, 10\} = \sim_{c_1} (10),\\
&\sim_{c_1} (01) = \{01, 11\} = \sim_{c_1} (11),\\
&\sim_{c_2} (00) = \{00, 01\} = \sim_{c_2} (01),\\
&\sim_{c_2} (01) = \{01, 11\} = \sim_{c_2} (11),\\
&V(0_{c_1}) = \{00, 01\}, \quad V(1_{c_1}) = S \setminus V(0_{c_1}),\\
&V(0_{c_2}) = \{00, 10\}, \quad V(1_{c_2}) = S \setminus V(0_{c_2}).
\end{aligned}$$

After Tandon's statement $M'^{(0)}$: The product update $M \otimes M^a$ yields

$$\begin{aligned}
&S'^{(0)} = \{(01, A), (10, A), (11, A), (00, B), (01, B), (10, B), (11, B)\},\\
&\forall a \in \{A, B\} :\\
&\quad \sim'^{(0)}_{c_1} (01, a) = \{(01, A), (11, A), (01, B), (11, B)\} = \sim'^{(0)}_{c_1} (11, a),\\
&\quad \sim'^{(0)}_{c_1} (10, a) = \{(10, A), (00, B), (10, B)\} = \sim'^{(0)}_{c_1} (00, B),\\
&\quad \sim'^{(0)}_{c_2} (10, a) = \{(10, a), (11, a)\} = \sim'^{(0)}_{c_2} (11, a),\\
&\quad \sim'^{(0)}_{c_2} (00, B) = \{(00, B), (01, B)\} = \sim'^{(0)}_{c_2} (01, B),\\
&\quad \sim'^{(0)}_{c_2} (01, A) = \{(01, A)\},\\
&V'^{(0)}(0_{c_1}) = \{(01, A), (00, B), (01, B)\}, \quad V'^{(0)}(1_{c_1}) = S'^{(0)} \setminus V'^{(0)}(0_{c_1}),\\
&V'^{(0)}(0_{c_2}) = \{(00, B), (10, A), (10, B)\}, \quad V'^{(0)}(1_{c_2}) = S'^{(0)} \setminus V'^{(0)}(0_{c_2}).
\end{aligned}$$

(ii) Is it possible for the children to ascertain whether they are muddy?

To answer this, we analyse what happens after each repetition round of the second announcement ϕ, *"Does any of you know whether you have mud on your own forehead?"*.

After the i-th repetition ($i \geq 1$) of ϕ (denoted ϕ_i), the restricted Kripke structure is $M'^{(i)} = M'^{(i-1)}|_{\phi_i} = \langle S'^{(i)}, \sim^{(i)}, V'^{(i)} \rangle$ with

$$\begin{aligned} S'^{(i)} &\coloneqq [\![\phi_i]\!] \coloneqq \{s \in S'^{(i-1)} : M'^{(i-1)}, s \models \phi_i\}, \\ \forall i \in \{1, \ldots, n\} &: \sim_{c_i}^{(i)} \coloneqq \sim_{c_i}^{(i-1)} \cap ([\![\phi_i]\!] \times [\![\phi_i]\!]), \\ V'^{(i)}(p) &\coloneqq V'^{(i-1)}(p) \cap [\![\phi_i]\!]. \end{aligned}$$

The encoding of ϕ and the formula are similar to (1.2(a4)) and (1.2(a5)) :

$$\begin{aligned} M'^{(i-1)}, s \models \phi_i \iff M'^{(i-1)}, s \models \bigwedge_{j=1}^{n} \neg K_{cj} 1_{cj} \wedge \neg K_{cj} 0_{cj} \\ \iff \bigwedge_{j=1}^{n} \Big(\neg\big(\forall t \in \sim_{c_j}'^{(i-1)} (s) : t \in V'^{(i-1)}(1_{c_j})\big) \wedge \\ \neg\big(\forall t \in \sim_{c_j}'^{(i-1)} (s) : t \in V'^{(i-1)}(0_{c_j})\big)\Big). \end{aligned} \tag{1.2(c3)}$$

Case $i = 1$ (first repetition): Define $m'_{ja1} = \{(l1r, A) : \forall l \in \{0\}^{j-1}, \forall r \in \{0\}^{n-j}\}$ (worlds in S' with only child j muddy) and $m'_{a1} = \bigcup_{j=2}^{n} m'_{ja1}$ (the set of worlds where exactly one child other than c_1 is muddy).

Observe that for $j = 1$,

$$\begin{aligned} &\forall (l1r, A) \in m'_{1a1} : \\ &\sim_{c_1}'^{(0)} ((l1r, A)) = \{(l1r, A), (l0r, B), (l1r, B)\} = \sim_{c_1}'^{(0)} ((l0r, B)), && (1.2(c4)) \\ &\forall (l1r, a), (l0r, b) \in S'^{(0)} \setminus m'_{a1} \text{ with } \forall l \in \{0, 1\}^{j-1}, \forall r \in \{0, 1\}^{n-j} : \\ &\sim_{c_1}'^{(0)} ((l1r, a)) = \{(l1r, a), (l1r, b), (l0r, a), (l0r, b)\} = \sim_{c_j}'^{(0)} ((l0r, b)). && (1.2(c5)) \end{aligned}$$

For all $j \geq 2$,

$$\begin{aligned} &\forall j \in \{2, 3, \ldots, n\} : \\ &\quad \forall (l1r, A) \in m'_{ja1} : \quad \sim_{c_j}'^{(0)} ((l1r, A)) = \{(l1r, A)\} \quad (\text{since } (l0r, A) \notin S'^{(0)}), && (1.2(c6)) \\ &\quad \forall (l1r, a), (l0r, a) \in S'^{(0)} \setminus m'_{a1} \text{ with } \forall l \in \{0, 1\}^{j-1}, \forall r \in \{0, 1\}^{n-j} : \\ &\quad \sim_{c_j}'^{(0)} ((l1r, a)) = \{(l1r, a), (l0r, a)\} = \sim_{c_j}'^{(0)} ((l0r, a)). && (1.2(c7)) \end{aligned}$$

From (1.2(c3)) and (1.2(c6)) , for any child c_j ($j \geq 2$) and any $s \in m'_{ja1}$, we have $M'^{(0)}, s \models K_{c_j} 1_{c_j}$ (because $(l1r, A) \notin V'^{(0)}(1_{c_j})$) and $M'^{(0)}, s \not\models K_{c_j} 0_{c_j}$ (because $(l1r, A) \notin V'^{(0)}(0_{c_j})$). Hence all worlds in m'_{ja1} fail to satisfy $\phi_1 (= \phi)$ and are eliminated from $M'^{(1)}$. This elimination is possible only because the world $(\{0\}^n, A)$ was already removed from $S^{(0)}$, which makes the equivalence classes of worlds in m'_{ja1} singletons.

For any world $s \in S^{(0)} \setminus m'_{a1}$, using (1.2(c4)) , (1.2(c5)) and (1.2(c7)) , we can argue that we will always find worlds $t_1, t_2 \in \sim_{c_j}'^{(0)} (s)$ for every $j \geq 1$ such that $t_1 \notin V'^{(0)}(1_{c_j})$ and $t_2 \notin V'^{(0)}(0_{c_j})$. Hence

$$\bigwedge_{j=1}^{n} \big(M'^{(0)}, s \not\models K_{c_j} 1_{c_j} \ \wedge \ M'^{(0)}, s \not\models K_{c_j} 0_{c_j}\big),$$

so they satisfy ϕ_1 and survive. Consequently, $S^{(1)} = S^{(0)} \setminus m'_{a1}$; i.e., all A**-worlds** (worlds of the form (s, A)) with **exactly one** muddy child (other than c_1) are removed after the first repetition, while A-worlds with two or more muddy children and all B-worlds survive.

Case $i = 2$ (second repetition): By the same reasoning as in case $i = 1$, because A-worlds with exactly one muddy child were eliminated after ϕ_1, t he equivalence classes for worlds with exactly two muddy children become singletons. Therefore, after the second repetition ϕ_2, all A-worlds with exactly two muddy children (other than c_1) are eliminated. Again, A-worlds where c_1 is muddy and all B-worlds survive.

We can frame this as a proof by induction: after the i-th repetition of ϕ, the surviving worlds are those A-worlds with **more than i muddy children** or with **child c_1 muddy**, together with **all B-worlds**. Hence, even after infinitely many repetitions of ϕ, A-worlds with child c_1 muddy and all B-worlds remain in the restricted Kripke structure. Other children can ascertain their muddy status only if Tandon announced A; otherwise, no one can deduce. Child c_1 can never deduce whether he has mud on his forehead in any case.

Intuitive verification: Since all children except child c_1 are attentive, they know for sure which announcement Tandon made: A ("At least one of you has mud on his forehead") or B (vacuous statement). Therefore, with each repetition they can deduce whether they are muddy (as in the previous cases). However, child c_1 is inattentive and never knows which announcement was made, so he can never deduce his own muddy status. In the case of a vacuous statement B, there is insufficient information about the true world, for any child to ever deduces anything.

(d)

We assume the same initial Kripke structure as in (1.2(a1)) . The only change is that the initial announcement ϕ_0 was *"At least one of you has mud on your forehead"*, but now it is changed to *"Child 1 has mud on his forehead"*. Accordingly, the encoding of the announcement becomes $\phi_0 = 1_{c_1}$.

After announcement $\phi_0 = 1_{c_1}$: The restricted Kripke structure $M^{(0)} = M|_{\phi_0}$ is defined as in (1.2(a2)) with $S^{(0)} \coloneqq [\![\phi_0]\!] = \{s \in S : M, s \models \phi_0\}$. For any world $s \in S$,

$$\begin{aligned} M, s \models \phi_0 &\iff M, s \models 1_{c_1} \\ &\iff s \in V(1_{c_1}). \end{aligned} \tag{1.2(d1)}$$

From (1.2(d1)) only the worlds in $V(1_{c_1})$ survive the announcement, hence

$$\begin{aligned} & S^{(0)} \coloneqq [\![\phi_0]\!] = V(1_{c_1}), \\ & \sim_{c_1}^{(0)} (1r) = \{1r\} \quad \text{for all } r \in \{0,1\}^{n-1}, \\ & \forall j \in \{2, \dots, n\}: \\ & \qquad \sim_{c_j}^{(0)} (1l1r) = \{1l1r, 1l0r\} = \sim_{c_j}^{(0)} (1l0r) \\ & \qquad \text{where } \forall l \in \{0,1\}^{j-1},\ \forall r \in \{0,1\}^{n-j-1}. \end{aligned} \tag{1.2(d2)}$$

The valuation function is restricted similarly.

Can others deduce whether they are muddy? To answer this, we check which worlds allow at least one child other than *child 1* to deduce his muddy status. For that, we restrict the Kripke structure against the repetitive announcement

$$\phi = \bigwedge_{j=2}^{n} \big(\neg K_{c_j} 1_{c_j} \wedge \neg K_{c_j} 0_{c_j}\big).$$

After the i-th repetition, the restricted structure is $M^{(i)} = M^{(i-1)}|_{\phi_i} = \langle S^{(i)}, \sim^{(i)}, V^{(i)}\rangle$. Worlds in which any other child knows his own state are eliminated by ϕ. Since the announcement is similar to that in (1.2(a5)) , the formula becomes:

$$\begin{aligned} M^{(i-1)}, s \models \phi_i \iff\ & M^{(i-1)}, s \models \bigwedge_{j=2}^{n} \big(\neg K_{c_j} 1_{c_j} \wedge \neg K_{c_j} 0_{c_j}\big) \\ \iff\ & \bigwedge_{j=2}^{n} \Big(\neg\big(\forall t \in \sim_{c_j}^{(i-1)} (s) : t \in V^{(i-1)}(1_{c_j})\big) \\ & \wedge\ \neg\big(\forall t \in \sim_{c_j}^{(i-1)} (s) : t \in V^{(i-1)}(0_{c_j})\big)\Big). \end{aligned} \tag{1.2(d3)}$$

Case $i = 1$ (first repetition): Analysing the equivalence relation $\sim^{(0)}$ from (1.2(d2)) , for any $s \in S^{(0)}$ we have:

$$\begin{aligned} & \forall j \in \{2, \ldots, n\} : \exists t_1, t_2 \in \sim_{c_j}^{(0)} (s) : t_1 \notin V^{(0)}(1_{c_j}) \ \wedge\ t_2 \notin V^{(0)}(0_{c_j}) \\ & \qquad \text{(choose } t_1 = 1l0r,\ t_2 = 1l1r \text{ with appropriate } l, r) \\ \iff & \forall j \in \{2, \ldots, n\} : \neg\big(\forall t \in \sim_{c_j}^{(0)} (s) : t \in V^{(0)}(1_{c_j})\big) \ \wedge\ \neg\big(\forall t \in \sim_{c_j}^{(0)} (s) : t \in V^{(0)}(0_{c_j})\big) \\ \iff & \bigwedge_{j=2}^{n} \Big(\neg(M^{(0)}, s \models K_{c_j} 1_{c_j}) \ \wedge\ \neg(M^{(0)}, s \models K_{c_j} 0_{c_j})\Big) \\ \iff & \bigwedge_{j=2}^{n} \big(M^{(0)}, s \models \neg K_{c_j} 1_{c_j} \wedge \neg K_{c_j} 0_{c_j}\big). \end{aligned} \tag{1.2(d4)}$$

Since every world $s \in S^{(0)}$ satisfies (1.2(d4)) , all worlds survive the announcement. Thus $S^{(1)} := [\![\phi_1]\!] = S^{(0)}$ and $M^{(1)} = M^{(0)}|_{\phi_1} = M^{(0)}$. This reflects that, even after the first repetition, no child other than *child 1* can deduce his muddy status.

Case $i = 2, \ldots$: The same pattern holds for all further repetitions of ϕ because the structure $M^{(i-1)}$ and the announcement ϕ remain identical each round. Hence,

$$\forall i \geq 2 : \quad M^{(i)} = M^{(i-1)} = M^{(0)}.$$

This shows mathematically that even after infinitely many repetitions of the announcement ϕ, no child other than *child 1* can deduce whether he is muddy.

Intuitive verification: The announcement *"Child 1 has mud on his forehead"* (ϕ_{c1}) initially conveys more information than the announcement *"At least one child has mud"* (ϕ_0). However, as repetitions of the second announcement (ϕ) proceed, ϕ_{c1} does not propagate any further information, whereas ϕ_0 propagates the information that **more than** $i-1$ children are muddy at round i. Since ϕ_0 unfolds additional information with each repetition, the children were able to deduce their own muddy status, which was not possible with ϕ_{c1}.

Problem 1.3

(a)

If we *consider* the *suits* of the cards, then there are $\binom{8}{2} \cdot \binom{6}{2} \cdot \binom{4}{2} = 2520$ possible worlds.

(b)

If we *ignore* the *suits* of the cards, then there are 19 possible worlds. Consider the distributions of *aces* among the hands of each player and the desk:

$$(2,2,0,0) \to \binom{4}{2} = 6$$
$$(2,1,1,0) \to \binom{4}{1} \cdot \binom{3}{2} = 12$$
$$(1,1,1,1) \to 1$$

Hence the total is $6 + 12 + 1 = 19$ possible worlds.

(c)

We can define the initial Kripke structure (ignoring the *suits*) as $M = \langle S, \sim, V \rangle$, such that the states are the distributions of *aces* to Aneeket, Raj, me, and the desk (respectively).

$$
\begin{gathered}
S = \{a_1a_2a_3a_4 : \forall i \in \{1,2,3,4\} : a_i \in \{0,1,2\} \wedge \sum_{i=1}^{4} a_i = 4\}, \\
\text{Agents} = \{a, r, y\}, \\
a_1a_2a_3a_4 \sim_a b_1b_2b_3b_4 \iff a_2 = b_2 \wedge a_3 = b_3, \\
a_1a_2a_3a_4 \sim_r b_1b_2b_3b_4 \iff a_1 = b_1 \wedge a_3 = b_3, \\
a_1a_2a_3a_4 \sim_y b_1b_2b_3b_4 \iff a_1 = b_1 \wedge a_2 = b_2, \\
\forall n \in \{0,1,2\} : \\
p_a^{(n)} = \{na_1a_2a_3 : \forall i \in \{1,2,3\} : a_i \in \{0,1,2\} \wedge \sum_{i=1}^{3} a_i = 4 - n\}, \\
p_r^{(n)} = \{a_1na_2a_3 : \forall i \in \{1,2,3\} : a_i \in \{0,1,2\} \wedge \sum_{i=1}^{3} a_i = 4 - n\}, \\
p_y^{(n)} = \{a_1a_2na_3 : \forall i \in \{1,2,3\} : a_i \in \{0,1,2\} \wedge \sum_{i=1}^{3} a_i = 4 - n\}.
\end{gathered}
\tag{1.3(c1)}
$$

(d)

I am holding **one ace and one eight**.

Intuitively: Since only two aces and two eights remain, the possible cards in my hand are one ace and one eight, two aces, or two eights. If I had two aces, then Raj would know he must have two aces; similarly, if I had two eights, then Aneeket would know he must have two aces. Hence the only possible choice is one ace and one eight.

Considering the Kripke structure: We encode the announcements as

$$
\phi_i = \bigwedge_{n=0}^{2} \neg K_i p_i^{(n)}
$$

for agent i being unable to determine his own cards. The restricted Kripke structure after ϕ_a is $M^{(1)} = M|_{\phi_a} = \langle S^{(1)}, \sim^{(1)}, V^{(1)} \rangle$, where $S^{(1)} = [\![\phi_a]\!] = \{s \in S : M, s \models \phi_a\}$.

$$\begin{aligned} M, s \models \phi_a \iff & M, s \models \bigwedge_{n=0}^{2} \neg K_a p_a^{(n)} \\ \iff & \bigwedge_{n=0}^{2} M, s \models \neg K_a p_a^{(n)} \\ \iff & \bigwedge_{n=0}^{2} M, s \models \neg\big(\forall t \in \sim_a (s) : t \in p_a^{(n)}\big) \end{aligned} \tag{1.3(c2)}$$

Define

$$a = \bigcup_{n \in \{0,2\}} \{a_1 n n a_4 \in S\}$$

representing the worlds where Raj and I both have either two aces or two eights. From (1.3(c2)) we observe that no world in a satisfies the formula. Intuitively, for any world in a, Aneeket would be able to know his cards, so all those worlds should be eliminated after his announcement. Thus $S^{(1)} = [\![\phi_a]\!] = S \setminus a$, and the corresponding edges are removed from the Kripke structure.

Similarly, for Raj's announcement we define

$$r = \bigcup_{n \in \{0,2\}} \{n a_2 n a_4 \in S^{(0)}\}$$

representing the worlds where Aneeket and I both have either two aces or two eights. The formula (1.3(c2)) is updated by replacing $\sim_a$ with $\sim_r^{(1)}$ and M with $M^{(1)}$. All these worlds are eliminated, giving $S^{(2)} = [\![\phi_r]\!] = S^{(1)} \setminus r$, with their corresponding edges removed.

(e)

Again, I am holding **one ace and one eight**.

Intuitively: Since three aces and one eight remain, the possible cards in my hand are one ace and one eight or two aces. Initially I would not know my cards in any case. If I had two aces, then Aneeket would not know his cards, but after Aneeket's announcement Raj (who can see my and Aneeket's cards) would know that he must have one ace and one eight. Because if Raj had two aces or two eights, then either I or Aneeket would have known our cards. Hence the only possible hand is one ace and one eight.

Considering the Kripke structure: The encoding of the announcements is the same as in part (d). Define

$$y = \bigcup_{n \in \{0,2\}} \{n n a_3 a_4 \in S\}$$

representing the worlds where Aneeket and Raj both have either two aces or two eights. The formula (1.3(c2)) is updated by replacing $\sim_a$ with $\sim_y$. With the same logic as above, for all worlds in y I would be able to know my cards, so they are eliminated (along with their corresponding edges), yielding $S^{(1)} = [\![\phi_y]\!] = S \setminus y$. In the same way, we can define sets a and r, and they will be eliminated after the announcements ϕ_a and ϕ_r respectively, with their corresponding edges removed.

Part-2

Problem 2.1

(a)

Both frameworks implement **Dynamic Epistemic Logic (DEL)**. They start with an epistemic model (a Kripke structure) representing the agents' knowledge and then update it when events occur. The differences lie in the kinds of events they support and the underlying data structures.

`demo_s5` – **Public announcements only**

Representation.

- **States:** any hashable objects (e.g., tuples, integers).
- **Agents:** identified by integers (e.g., $a = 0$, $b = 1$).
- **Valuation:** a mapping from states to a set of propositional atoms. The logic is **2-valued** (True/False) – every proposition is either true or false at every world.
- **Accessibility:** stored as **partitions** (equivalence relations). For each agent, a list of blocks (each block is a list of states). This enforces S5 properties (reflexive, transitive, symmetric).
- **Formulas:** `Top`, `Info(world)` (true only at that world), `Ng` (negation), `Conj`, `Disj`, `Kn(agent,form)` (knowledge).

Event handling. Only **public announcements** are supported via the function `upd_pa(model, form)`. It:

1. Keeps only those states where the announced formula is true.
2. Restricts each agent's partition to the remaining states (drops worlds and updates blocks).
3. Keeps only actual worlds that survive.

No other actions (private messages, factual changes, etc.) can be expressed.

Solving a DEL problem. In `demo_s5`, the initial epistemic model M must be **explicitly constructed** to encode all relevant information of the puzzle. For Cheryl's birthday, M contains exactly the ten candidate dates, with Albert's relation grouping by month and Bernard's relation grouping by day. The actual world is chosen from among these states. There is no way to "set" the real situation later – everything is given into the initial model. A DEL problem in this framework is specified by:

1. An initial epistemic model M (states, partitions for each agent, valuation, actual worlds).
2. A sequence of public announcements $\phi_1, \phi_2, \dots, \phi_n$.

The solution proceeds by repeatedly applying `upd_pa`:

$$M^{(0)} = M, \qquad M^{(i+1)} = \texttt{upd\_pa}(M^{(i)}, \phi_{i+1}).$$

After each announcement, the model shrinks: worlds where the announced formula is false disappear. The final model M_n contains exactly those worlds that are consistent with all announcements. If the puzzle asks "what is the real world?" (e.g., Cheryl's birthday), the answer is the unique actual world in M_n (if exactly one remains). If the puzzle asks "does agent a know ψ?", one checks $\texttt{is\_true}(M_n, \texttt{Kn}(a, \psi))$.

`demo_light` – Action models with vocabulary change

Representation.

- **States:** integers (after bisimulation) or any hashable type.
- **Agents:** same as above.
- **Vocabulary (`voc`):** an explicit list of propositional atoms that "exist". Truth evaluation is **3-valued** (True / False / None) – a proposition not in `voc` is undefined. This allows modelling ignorance about the existence of a fact.
- **Valuation:** mapping from states to the set of atoms true there (only atoms from `voc` appear).
- **Accessibility:** stored as a **list of triples** `(ag, state1, state2)` – the states(`state1,state2`) that agent(`ag`) can't distinguish.
- **Formulas:** `Top`, `Prp(atom)`, `Neg`, `Conj`, `Disj`, `K(agent,form)`, `CK(agents,form)` (common knowledge).

Event handling – Action Models with Vocabulary Change (ACM). An **ACM** is a generalisation of a public announcement. It consists of:

- `states`: a set of event indices (e.g., $0, 1, 2$).
- `pre`: a precondition formula for each event.
- `subst`: for each event, a list of pairs `(proposition, formula)`. When the event occurs, each proposition is **replaced** by the given formula (evaluated in the original world). This models **factual change** (e.g., flipping a switch, painting a wall).
- `rels`: a list of triples `(agent, event1, event2)` describing which events an agent cannot distinguish.
- `actual`: the event(s) that actually happen.

Product update `upc(m, facm)` (or `upd` followed by bisimulation) computes the new epistemic model:

1. New states are pairs `(world, event)` where the precondition of the event holds at the world.
2. New vocabulary = old vocabulary $\cup$ all propositions appearing in any substitution of the ACM.
3. New valuation: for each `(w,s)`, a proposition p is true iff its substitution (or p itself if no substitution) evaluates to true at w ($m, w \models p$ or $sub(p)$).
4. New relations: `(w1,s1)` and `(w2,s2)` are related for agent ag iff $w1$ and $w2$ were related in m and $s1$ and $s2$ are related in the $facm$.
5. Actual worlds: pairs of actual world and actual event.

High-level constructors (all return ACMs) include:

- `public(form)` – single event, precondition `form`, no substitution. (Public announcement)
- `group_m(group, form)` – group learns `form`; outsiders cannot distinguish it from `Top`.
- `message(agent, form)` – private message to one agent.
- `test(form)` – silent test (no one observes the restriction).
- `info(group, form)` – group learns **whether** `form` is true or false (two events).
- `p_change(substitution)` – public factual change (all agents see the substitution).
- `perception(agent, form, group)` – agent and group observe the truth value of `form` (no change).
- `perceived_change(agent, prp, form, group)` – agent and group observe the old truth value of `prp` and simultaneously change it to `form`.

Solving a DEL problem. In `demo_light`, the initial model does **not** need to be hand-crafted for the specific puzzle. Instead, one typically starts from a **blissful-ignorance model** generated by `initM(agents, props)`. This model contains all logically possible worlds (all truth assignments to the given propositions). The real situation is then imposed **dynamically** using ACM actions.For example:

- `test(actual_situation)` restricts the model to worlds where the actual situation holds, but no agent observes this restriction – it is a silent, unobserved factual setting.
- Further actions (`info`, `message`, `group_m`, `p_change`, `perception`, etc.) refine the model by adding private observations, factual changes, or vocabulary modifications.

Thus, a DEL problem is specified by:

1. An initial epistemic model M (states, relations, valuation, actual worlds).
2. A sequence of actions $\alpha_1, \alpha_2, \ldots, \alpha_n$, where each α_i is an ACM (or a high-level constructor that produces one).

The solution proceeds by repeatedly applying product update:

$$M^{(0)} = M, \qquad M^{(i+1)} = \mathtt{upd}(M^i, \alpha_{i+1})$$

After each action, the model may grow (because of pairs with different events) or shrink (because of preconditions). The final model $M^{(n)}$ represents the epistemic situation after all actions. Queries (e.g., "does agent a know ψ?", "what is the value of proposition p?") are answered by evaluating the relevant formula in $M^{(n)}$ using `is_true`.

(b)

The extra representational power of `demo_light` comes from two key innovations:

1. **Explicit vocabulary and 3-valued logic.**
   - `demo_s5` uses a fixed set of propositions; every proposition is either true or false at every world. There is no notion of an "undefined" proposition.
   - `demo_light` maintains an explicit list `voc` of atoms that exist. A proposition not in `voc` evaluates to `None` (undefined). This allows modelling situations where agents are not even aware of a certain fact, or where new propositions are introduced dynamically (e.g., through an ACM substitution).
2. **Action models with substitution (ACM).**
   - `demo_s5` can only perform **public announcements** – a single event that all agents know has occurred, and which only removes worlds.
   - `demo_light` supports **any action model**, including:
     - **Private and group announcements** (e.g., `message`, `group_m`, `info`) – where different agents have different access to the events.

  - **Silent tests** (`test`) – the world changes but no agent observes it.
  - **Factual changes** (`p_change`, `unobserved_change`, `perceived_change`) – the truth value of propositions can be altered, not just eliminated.
  - **Perception changes** – an agent observes the truth of a formula (`perception`) or observes a factual change and thereby learns the old value (`perceived_change`).
    For example, if an agent sees a light switch being flipped, they learn that the light was previously in the opposite state.
- Because public announcements are a special case of an ACM (single event with no substitution), `demo_light` can simulate every scenario that `demo_s5` can, but the converse is false.

In summary, `demo_light` is strictly more expressive: it can model public announcements, private messages, factual changes, unobserved changes, perception changes, and combinations thereof – all within a unified framework of action models with vocabulary change. `demo_s5` is limited to public announcements only, making it suitable for puzzles like Cheryl's birthday but inadequate for scenarios involving private information or world-altering events.

(c)

(i) Can the Sally–Anne test be modeled using `demo_s5`? No. The `demo_s5` framework has two fundamental limitations that make it incapable of modeling the Sally–Anne scenario:

1. **Only public announcements.** `demo_s5` supports only one type of event: a truthful public announcement that all agents hear and know that everyone hears. In the Sally–Anne test, Anne moves the toy **privately** while Sally is out of the room. Sally does not observe the move, so her belief remains outdated. This is a private, unobserved action. Public announcements cannot capture such asymmetric information updates.

2. **No factual change.** In `demo_s5`, the valuation (which propositions are true at each world) is fixed at the start and never changes. The only operation is world elimination. The Sally–Anne test requires a **factual change**: the toy's location changes from Sally's basket to Anne's box. This cannot be represented by simply removing worlds; the truth value of the proposition "toy in Sally's basket" must become false, and "toy in Anne's box" must become true. `demo_s5` has no mechanism for substitution or valuation update.

One might attempt to encode both possible locations in the initial model and then use a public announcement that "Anne moves the toy". However, that announcement would be heard by Sally, making her aware of the move and destroying the false belief that is essential to the test. Therefore, `demo_s5` cannot model the Sally–Anne test.

(ii) Can it be modeled using `demo_light`? Yes.

`demo_light` can model the test using action models with vocabulary change (ACM). The implementation follows these steps (see `sally_anne.py`):

1. **Agents and propositions:** Sally (a) and Anne (b); $P(1)$ = "toy in Sally's basket", $P(2)$ = "toy in Anne's box".

2. **Initial model:** Start from a blissful-ignorance model (`initM([sally, anne], [P(1), P(2)])`) containing all $2^2 = 4$ worlds. Silently set the actual world (toy in Sally's basket) using `test(Conj([in_sally, Neg(in_anne)]))`.

3. **Private move action (ACM):** Construct an action model with two events: 0 (Anne moves the toy) and 1 (no move).

   - Preconditions: event 0 requires `in_sally`; event 1 requires $\top$.
   - Substitutions: event 0 maps $P(1)$ to `False` and $P(2)$ to `True`; event 1 does nothing.
   - Accessibility: Anne distinguishes 0 from 1; Sally cannot distinguish them (edges $(0, 1)$ and $(1, 0)$).
   - Actual event: 0 (Anne moves the toy).

   The ACM is applied with `upd(m1, private_move())`.

4. **Verification:** After the update, evaluate:

   - $K(\text{Sally}, P(2))$ – Sally does know the toy is in Anne's box. (False)
   - $K(\text{Anne}, P(2))$ – Anne knows the true location. (True)
   - Sally considers it possible that the toy is still in her basket (i.e., $\neg K(\text{Sally}, \neg P(1))$), capturing her false belief. (True)

The program outputs the expected truth values, confirming that `demo_light` successfully models the false-belief scenario.